TE 10 Mode

Aim: To simulate and compare the TE_{10} and TM_{11} modes in a WR-90 rectangular waveguide using HFSS, determine their cutoff frequencies, propagation constants, and field distributions, and verify why TE_{10} propagates while TM_{11} does not at 10 GHz.

Software required:
Matlab

Program:

```
clc;
clear;
close all;

%% WR-90 Waveguide Parameters
a = 22.86e-3;       % Width in meters
b = 10.16e-3;       % Height in meters
f = 10e9;           % Operating frequency = 10 GHz
c = 3e8;            % Speed of light
mu0 = 4*pi*1e-7;
eps0 = 8.854e-12;

k0 = 2*pi*f/c;

%% Cutoff Frequencies
fc_TE10 = (c/2)*sqrt((1/a)^2);
fc_TM11 = (c/2)*sqrt((1/a)^2 + (1/b)^2);

%% Propagation Constants
beta_TE10 = 2*pi*sqrt((f/c)^2 - (fc_TE10/c)^2);

% TM11 is below cutoff at 10 GHz
if f > fc_TM11
    beta_TM11 = 2*pi*sqrt((f/c)^2 - (fc_TM11/c)^2);
else
    alpha_TM11 = 2*pi*sqrt((fc_TM11/c)^2 -
(f/c)^2);
    beta_TM11 = 0;
end

%% Display Numerical Results
fprintf('OPERATING FREQUENCY = %.2f
GHz\n', f/1e9);

fprintf('\nTE10 MODE\n');
fprintf('Cutoff frequency = %.4f GHz\n',
fc_TE10/1e9);
fprintf('Propagation constant beta = %.3f rad/m\n',
beta_TE10);

fprintf('\nTM11 MODE\n');
fprintf('Cutoff frequency = %.4f GHz\n',
fc_TM11/1e9);

if f > fc_TM11
    fprintf('Propagation constant beta = %.3f
rad/m\n', beta_TM11);
else
    fprintf('Below cutoff: beta is imaginary for the
propagation solution.\n');
    fprintf('Evanescent attenuation constant alpha
```

```
%% Plot TE10 Electric Field
figure('Name','TE10 E-field');
surf(X*1000,Y*1000,Ey_TE10);
shading interp;
colorbar;
xlabel('x (mm)');
ylabel('y (mm)');
zlabel('Normalized E_y');
title('TE10 Electric Field Distribution');
view(2);
axis tight;

%% Plot TE10 Magnetic Field
figure('Name','TE10 H-field');
surf(X*1000,Y*1000,Hz_TE10);
shading interp;
colorbar;
xlabel('x (mm)');
ylabel('y (mm)');
zlabel('Normalized H_z');
title('TE10 Magnetic Field Distribution');
view(2);
axis tight;

%% Plot TM11 Electric Field
figure('Name','TM11 E-field');
surf(X*1000,Y*1000,Ez_TM11);
shading interp;
colorbar;
xlabel('x (mm)');
ylabel('y (mm)');
zlabel('Normalized E_z');
title('TM11 Electric Field Distribution');
view(2);
axis tight;

%% Plot TM11 Magnetic Field
figure('Name','TM11 H-field');
surf(X*1000,Y*1000,Hx_TM11);
shading interp;
colorbar;
xlabel('x (mm)');
ylabel('y (mm)');
zlabel('Normalized H_x');
title('TM11 Magnetic Field Distribution');
view(2);
axis tight;

%% Mode Comparison
figure('Name','Mode Comparison');
bar([fc_TE10/1e9,fc_TM11/1e9]);
set(gca,'XTickLabel',{'TE10','TM11'});
```

```
= %.3f Np/m\n', ...
      alpha_TM11);
end

%% Field Distribution Grid
Nx = 100;
Ny = 60;
x = linspace(0,a,Nx);
y = linspace(0,b,Ny);
[X,Y] = meshgrid(x,y);

%% TE10 Electric Field
% Transverse electric field Ey
Ey_TE10 = sin(pi*X/a);

%% TE10 Magnetic Field
% Hz and Hx spatial patterns (normalized)
Hz_TE10 = cos(pi*X/a);
Hx_TE10 = sin(pi*X/a);

%% TM11 Electric Field
% Longitudinal electric field Ez
Ez_TM11 = sin(pi*X/a).*sin(pi*Y/b);

% Transverse electric field components
(normalized)
Ex_TM11 = (pi/a)*cos(pi*X/a).*sin(pi*Y/b);
Ey_TM11 = (pi/b)*sin(pi*X/a).*cos(pi*Y/b);

%% TM11 Magnetic Field (normalized spatial
pattern)
% Hz = 0 for TM modes
% Transverse H-field components
Hx_TM11 = sin(pi*X/a).*cos(pi*Y/b);
Hy_TM11 = cos(pi*X/a).*sin(pi*Y/b);
```

```
ylabel('Cutoff Frequency (GHz)');
title('Cutoff Frequency Comparison');
grid on;

%% Mode Decision
fprintf('\nMODE DECISION AT 10 GHz:\n');

if f > fc_TE10
   fprintf('TE10 propagates.\n');
else
   fprintf('TE10 is below cutoff.\n');
end

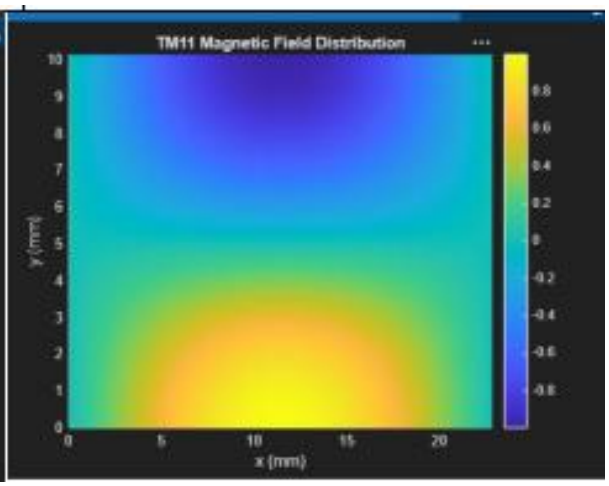
if f > fc_TM11
   fprintf('TM11 propagates.\n');
else
   fprintf('TM11 is below cutoff and is
evanescent.\n');
end
```

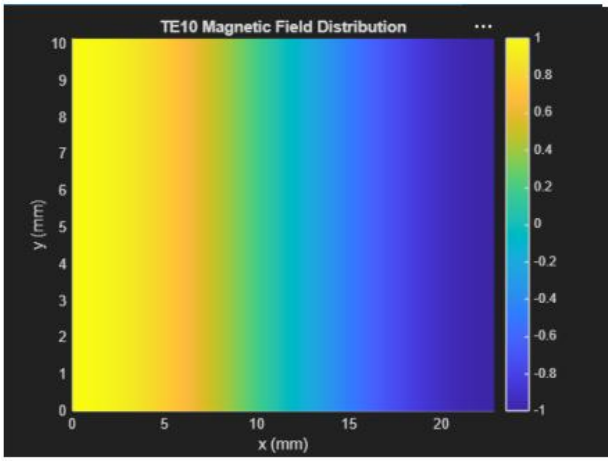
OUTPUT:

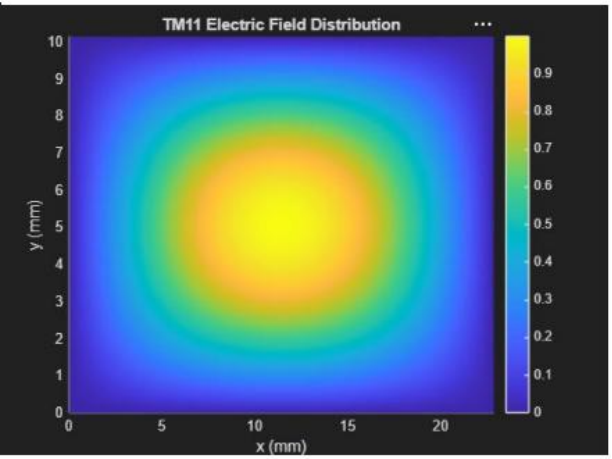
```
TM11 MODE
Cutoff frequency = 16.1563 GHz
Below cutoff: beta is imaginary for the propagation solution.
Evanescent attenuation constant alpha = 265.769 Np/m

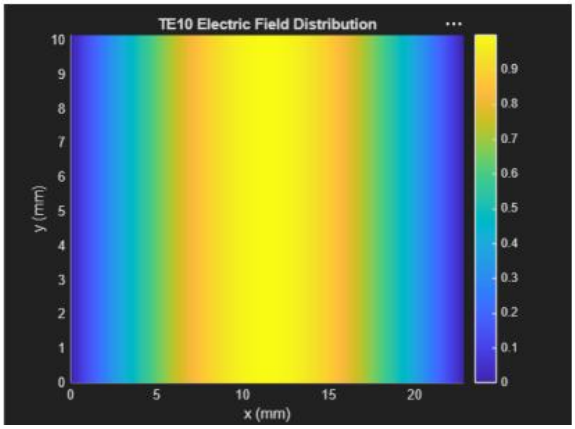
MODE DECISION AT 10 GHz:
TE10 propagates.
TM11 is below cutoff and is evanescent.
```






TE10 Magnetic Field Distribution
y (mm)
x (mm)


TM11 Electric Field Distribution
y (mm)
x (mm)


TE10 Electric Field Distribution
y (mm)
x (mm)